

Q4 The update report

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1 Introduction

The new update refactors the codebase to an object-oriented architecture, replaces NumPy/pandas-based vectorization with TensorFlow, with proper docstrings and comments(for more complex lines) throughout.

In this document, we address the questions from previous email and include all new updates in this single file. More specifically, we first explain why methods like VFI is more certain to be correct, and then solve the VFI benchmark solution for the comparison with the DL solution. Finally, we show side-by-side plots for better comparison between VFI and DL methods.

2 Responses to Questions

2.1 Further experiment and benchmarking

Question

- (a) Is there a way to test that your trained deep learning function results in a correct solution? For example, can you find a benchmark that is more certain to be correct and compare the deep learning solutions to the benchmark?
Hint: in the low-dimensional case, value function iteration (VFI) and policy function iteration (PFI) still work. Are the solutions from VFI and PFI “more certain to be correct” and suitable for being used as a benchmark?

- (b) If yes, what property or theorem guarantees that?
- (c) If there is a benchmark value function and policy function, how do you compare the deep learning result with the benchmark result?
- (d) If there is no benchmark, how would you show the effectiveness of your solution? Will presenting ergodic sets as the illustration help?

Response

- (a) Yes, value function iteration (VFI) and policy function iteration (PFI) can be used as benchmarks. They are indeed “more certain” to be correct in low-dimensional settings. We shall mainly focus on value function iteration on this update report.
- (b) This is guaranteed by the contraction mapping theorem on the Bellman operator[1, 2]. In §3.1.2, Strebulaev and Whited describe a discrete-state value function iteration algorithm:
 - Choose finite grids $K = \{k_1, \dots, k_{n_k}\}$ and $Z = \{z_1, \dots, z_{n_z}\}$.
 - Approximate the Markov process for z with a finite-state Markov chain on Z , with transition matrix P , where $P_{ij} \equiv \Pr(z' = z_j \mid z = z_i)$.
 - Initialize a guess for the value function on the grid, $V^{(0)}(k_i, z_j)$ (for example, zeros or the value with no adjustment of capital).
 - Iterate on the Bellman operator: given $V^{(m)}$, compute the right-hand side of the discrete Bellman equation for every (k_i, z_j) and all $k' \in K$, set $V^{(m+1)}(k_i, z_j)$ equal to the resulting maximum, and repeat until $V^{(m)}$ converges.
 - Record, for each state (k_i, z_j) , the maximizing choice of next-period capital, which is the discrete-state policy function.
 - With the converged policy function g , simulate the model forward: start from an initial (k_0, z_0) , draw a sequence for z_t from P , and update $k_{t+1} = g(k_t, z_t)$ to obtain artificial data for investment, capital, and cash flows.

In this discretized problem, the state space $S \equiv K \times Z$ is finite and a value function is simply a vector in $R^{|S|}$. The discrete Bellman operator is

$$(TV)(k, z) = \max_{k' \in K} \left\{ e(k, k', z) + \beta \sum_{z' \in Z} V(k', z') P(z, z') \right\}. \quad (1)$$

Let $\|\cdot\|_\infty$ denote the sup norm over the finite set S :

$$\|V\|_\infty \equiv \max_{(k, z) \in S} |V(k, z)|. \quad (2)$$

Fix finite grids K and Z , a transition matrix P on Z , and $\beta \in (0, 1)$. Then the discrete Bellman operator in Eq. (1) is a contraction on $(R^{|S|}, \|\cdot\|_\infty)$ with modulus β :

$$\|TV - TW\|_\infty \leq \beta \|V - W\|_\infty \quad \text{for all } V, W \in R^{|S|}. \quad (3)$$

Consequently, T has a unique fixed point V^* on the grid, and value function iteration $V_{n+1} = TV_n$ converges to V^* from any initial guess V_0 .

Fix $(k, z) \in S$. Let $k'_V \in \arg \max_{k' \in K} \{\dots\}$ be an optimizer in Eq.(1) under V . Then

$$(TV)(k, z) = e(k, k'_V, z) + \beta \sum_{z' \in Z} V(k'_V, z') P(z, z'). \quad (4)$$

Evaluating TW at the *same* feasible control k'_V yields

$$(TW)(k, z) \geq e(k, k'_V, z) + \beta \sum_{z' \in Z} W(k'_V, z') P(z, z'). \quad (5)$$

Subtracting gives

$$(TV)(k, z) - (TW)(k, z) \leq \beta \sum_{z' \in Z} (V(k'_V, z') - W(k'_V, z')) P(z, z') \quad (6)$$

$$\leq \beta \|V - W\|_\infty, \quad (7)$$

since $P(z, \cdot)$ is a probability distribution. Swapping the roles of V and W gives the reverse inequality, hence

$$|(TV)(k, z) - (TW)(k, z)| \leq \beta \|V - W\|_\infty \quad \forall (k, z) \in S. \quad (8)$$

Taking the maximum over S yields the contraction inequality. The fixed-point existence/uniqueness and VFI convergence then follow from the contraction mapping theorem.

More specifically, the Bellman operator is a contraction (and so VFI converges) if roughly (see [2]):

Discounting requires $0 < \beta < 1$. In our model, since $\beta = 1/(1+r)$ and $r > 0$, the condition is satisfied.

Boundedness In the numerical implementation, the state space is truncated by putting (k, z) on finite grids. Thus, the state space is finite and all payoffs are automatically bounded.

Feasible set and continuity For each (k, z) , the firm chooses next period's capital k' . Since the feasible set for k' is taken to be within an interval $[k'_{\min}, k'_{\max}]$, and the per-period payoff $e(k, k', z)$ is continuous in (k, k', z) , these satisfy the kinds of conditions used in the contraction theorem.

Markov shocks It requires z follows a Markov process with a well-defined transition kernel, which is exactly the assumption in our model.

Thus, the Bellman operator is a contraction and our VFI would converge to unique value from any initial guess and the convergence is geometric at rate β .

- (c) If there are other methods (such as VFI or PFI) that can serve as a benchmark for the DL method, we would compare the deep learning result with the benchmark result in the following aspects:

Policy-level agreement The DL policy $\iota^{DL}(k, z)$ should be uniformly very close to the VFI policy $\iota^{VFI}(k, z)$, especially on and near the ergodic set.

Value-level agreement The DL-implied value function $V^{DL}(k, z)$ should be very close to $V^{VFI}(k, z)$.

Simulation-level agreement If we simulate long panels under each policy, all economically relevant moments and distributions (means, variances, ACs, etc.) should line up within certain tolerances.

Regime behavior In specific regions (e.g. low k / low z ; high k / high z) the qualitative patterns and "regimes" match.

- (d) If there is no benchmark, we may show the effectiveness of the solution as follows:

Analytical solution check We can start from special cases where we know the answer in closed form, and see if our method reproduces them. For example, when there is no adjustment cost, we are able to obtain the analytical solution. By solving the model using our DL method in these special limits we can check whether our results reduce to those limits.

Bellman and Euler residuals check For each grid point (k, z) , we can compute the Bellman residual (difference between LHS and RHS of Bellman equation) and compute summary statistic such as sup-norm or L^2 norm. We can also check Euler-equation (FOC) errors along simulated paths. Using our solved policy, we simulate a long history $\{k_t, z_t\}_{t=0}^T$ and then compute the Euler residuals ϵ_t and compute for example the mean, max, 95th percentile etc.

Economic sanity checks We may check if the DL solution makes economic sense and replicates the paper's qualitative findings. For example, check that investment increases in the productivity shock z etc., or see if we can reproduce comparative statics.

Presenting ergodic sets as an illustration may help confirm that our reported Euler-equation errors are computed over the region visited under the learned policy, but may not related directly to the effectiveness of solution.

2.2 Boundary Condition

Question

Can FB function in Maliar(2021) be applied to the case in section 3.6 of Strebulaev(2012). If not, would you like to elaborate more by comparing equation (26) in Maliar(2021) and the case in section 3.6 of Strebulaev(2012)?

Response

We believe the FB construction in Maliar(2021) *cannot* be mechanically “dropped into” the model of Section 3.6 in Strebulaev and Whited (2012), because the nature of the nonlinearity is different: a local Kuhn–Tucker condition in the former versus a global optimal–stopping (limited–liability) problem in the latter.

FB in Maliar et al. (2021) and equation (26)

In the one–agent consumption–saving problem with a borrowing constraint, Maliar et al. consider the Bellman equation (their eq. (26))

$$V(y, w) = \max_{c, w'} \left\{ u(c) + \beta E[V(y', w')] \right\},$$

subject to the static constraint

$$c \leq w.$$

Writing the slack as $A \equiv w - c \geq 0$ and denoting by H the (appropriately normalized) Lagrange multiplier / Euler residual, the Kuhn–Tucker conditions can be cast as

$$A \geq 0, \quad H \geq 0, \quad AH = 0.$$

Maliar et al. work with unit–free variables

$$a \equiv 1 - \frac{c}{w}, \quad h \equiv 1 - \frac{\beta r E[u'(c')]}{u'(c)},$$

so that the Kuhn–Tucker system is

$$a \geq 0, \quad h \geq 0, \quad ah = 0.$$

The Fischer–Burmeister function

$$\Phi_{\text{FB}}(a, h) \equiv a + h - \sqrt{a^2 + h^2}$$

has the key property

$$\Phi_{\text{FB}}(a, h) = 0 \quad \Longleftrightarrow \quad a \geq 0, \quad h \geq 0, \quad ah = 0,$$

while being continuously differentiable in (a, h) . Thus the nonsmooth complementarity conditions are replaced by the single smooth equation $\Phi_{\text{FB}}(a, h) = 0$, and Maliar et al. minimize $\Phi_{\text{FB}}(a, h)^2$ (together with other residuals) in their deep-learning objective. Importantly, this use of FB is *local and intraperiod*: for a given state (y, w) one is smoothing a static Kuhn–Tucker system associated with the borrowing constraint in the inner maximization of the Bellman equation.

Section 3.6 of Strebulaev and Whited (2012)

In contrast, Section 3.6 in Strebulaev and Whited (2012) introduces risky debt via endogenous default in a discrete-time investment model. In simplified notation, their Bellman equation (their eq. (3.28)) can be written as

$$V(k, b, z) = \max \left\{ 0, \max_{k', b'} [e(k, k', b, b', z) + \eta(e) + \frac{1}{1+r} E[V(k', b', z')]] \right\}.$$

Here k is capital, b is debt, z is the shock, and $e(\cdot)$ collects current-period cash flows (after investment and financing). The outer $\max\{\cdot, 0\}$ implements limited liability and the default decision: equityholders either continue (value given by the inner maximization) or default and obtain value 0.

Crucially, the interest rate on debt, $\tilde{r}(z, k', b')$, is itself determined by a lender zero-profit condition (their eq. (3.27)) that depends on the set of future states in which $V(k', b', z') = 0$ (default). Thus this is an *optimal stopping problem* with an additional equilibrium pricing condition for risky debt, not a simple intraperiod inequality on a control variable.

Why FB cannot be applied directly

The FB construction in Maliar et al. hinges on having an explicit pair (a, h) of

(slack in constraint, multiplier / Euler residual)

satisfying a Kuhn–Tucker complementarity system at each state. In Section 3.6 of Strebulaev and Whited, there is no constraint of the form $g(\text{state}, \text{control}) \geq 0$ on the controls with an associated multiplier. Instead, the central nonsmooth element is the *default / continuation* decision embodied in

$$V(k, b, z) = \max\{0, TV(k, b, z)\},$$

where TV is the continuation operator given by the inner maximization and expectation. The endogenous risky interest rate further couples this operator to the entire future default region. Therefore, one cannot simply take the FB function as used for $c \leq w$ in Maliar (2021) and insert it into the equations of Section 3.6; the economic objects and the structure of the nonsmoothness are different.

2.3 Results and Visualization

Question

To improve interpretability, start with side-by-side figures (e.g., DL vs VFI policy) and propose any additional plots you believe would highlight convergence, accuracy, and regime behavior.

Response

Convergence diagnostics. The DL Euler-equation training curve shows a stable and monotone decline in the loss on a log scale after an initial transient phase, indicating that stochastic gradient updates are successfully reducing Euler-equation errors over time. On the other hand, the VFI sup-norm difference also decreases steadily, with near-linear behavior in log scale, which is the expected pattern for a contraction-type fixed-point iteration. Near the end, the curve exhibits a plateau around 10^{-4} followed by a transition to numerical zero; this should be interpreted as a finite-precision effect as the implementation uses `float32`, the effective resolution of updates is limited by machine precision.(Fig.1).

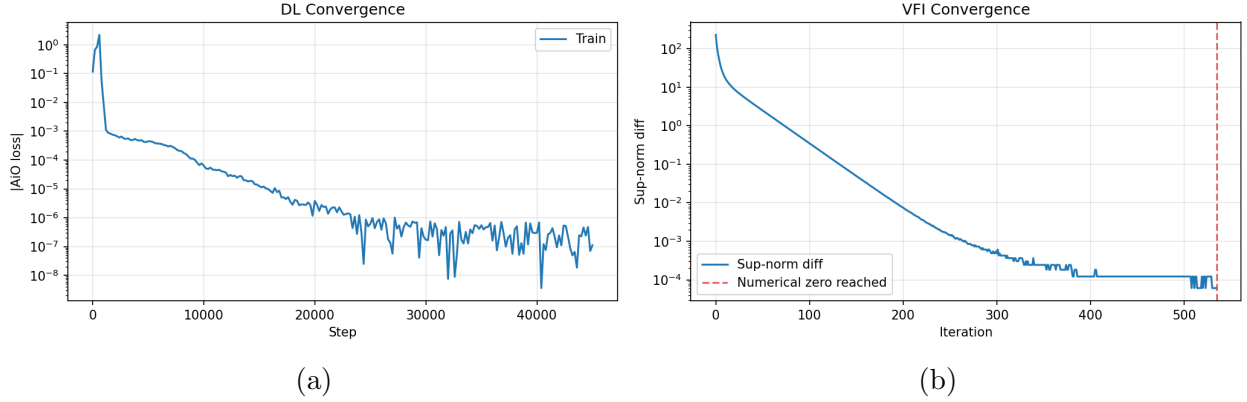


Figure 1: Convergence plots for both DL and VFI methods

Policy-level agreement We compute the VFI policy on a (k, z) grid and evaluate the trained DL policy on the same grid; the resulting policy surfaces $\iota^{VFI}(k, z)$ ($\iota = I/k$) and $\iota^{DL}(k, z)$ are plotted as heatmaps along with their difference(Fig.2). We also plot one-dimensional slices at low-, mid-, and high- z grid points to visualize cross-sections(Fig.3). The heatmaps and slices show that the DL policy is nearly indistinguishable from the VFI policy across the grid, with small and tightly centered differences; any residuals are largest at the edges of the grid where interpolation and discretization effects are most pronounced.

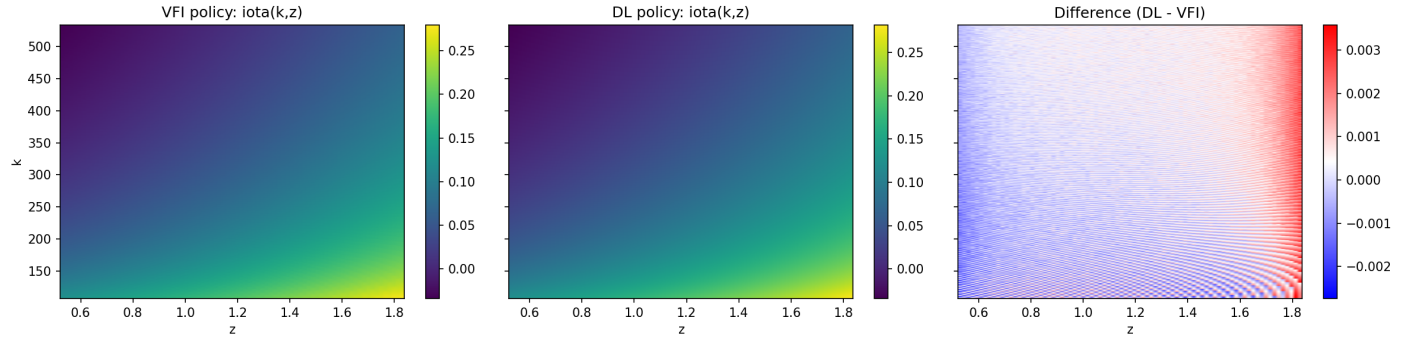


Figure 2: Policy comparison

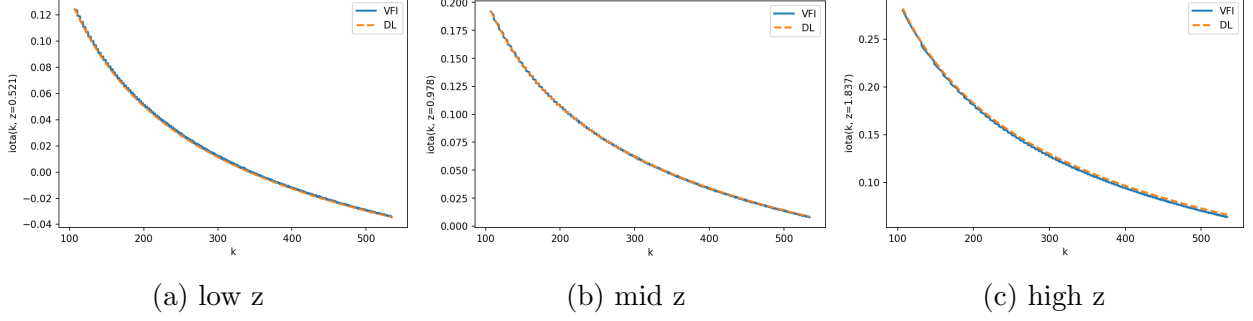


Figure 3: Slices for different fixed $z(\iota$ vs k)

Value-level agreement. We evaluate the DL policy value function $V^{DL}(k, z)$ by iterating on the Bellman equation with the DL policy held fixed, using the same transition matrix and grid as the VFI solution. We then plot $V^{VFI}(k, z)$, $V^{DL}(k, z)$, and their difference. The value-function plots are visually nearly identical, with small differences(Fig.4).

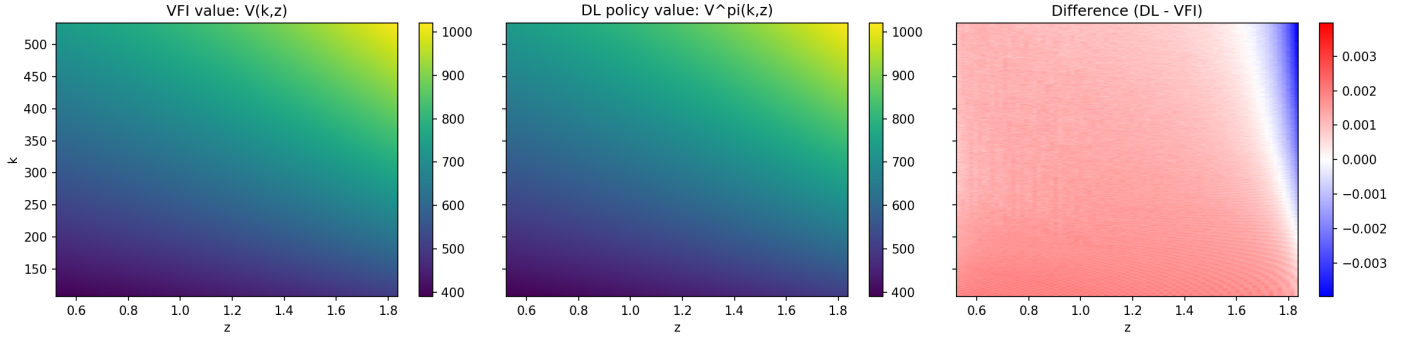


Figure 4: Value comparison

Simulation-level agreement. We simulate long on-policy panels for each policy after burn-in, using $T = 20,000$ time steps across 256 parallel paths. From these panels we compute the main ergodic moments—means, variances, and autocorrelations at lags 1 and 4 for k , $I = \iota k$, and $y = zk^\theta$. The resulting bar plots show the DL and VFI moments lining up closely across all variables and lags(Fig.5). Additional stationary distribution plots for k and ι provide shape-level confirmation(Fig.6). Overall, the simulation evidence indicates that the DL policy reproduces the VFI economy’s long-run distribution and persistence structure.

Regime behavior. We study regime behavior by simulating long on-policy samples under the DL and VFI policies (burn-in of 5,000 steps, then $T = 50,000$ total observations collected from 512 parallel paths). We form joint (k, z) regimes using empirical quantile bins: edges are the 0, 1/3, 2/3, and 1 quantiles of the pooled (k, z) samples, yielding a 3×3 grid of regimes. Within each regime we compute (i) the mean investment rate ι , (ii) its standard deviation, and (iii) the regime share (fraction of simulated states that fall in that bin). We then plot three heatmaps for each of these statistics: VFI, DL, and their difference

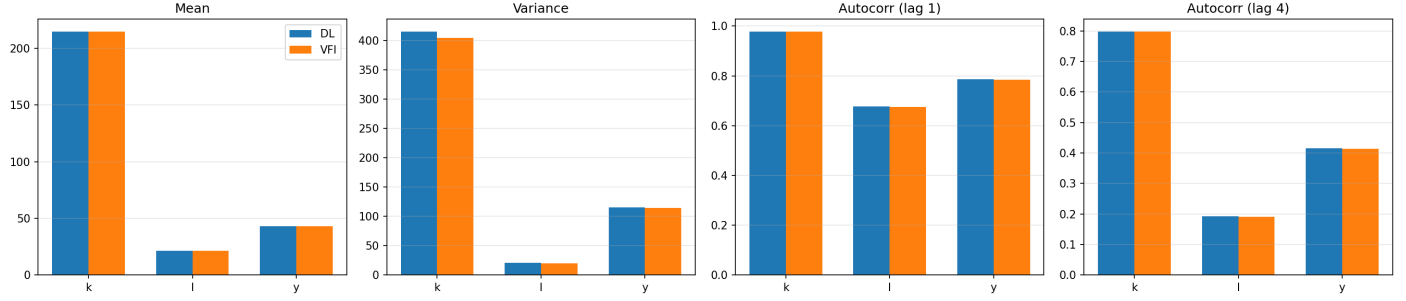


Figure 5: Mean, variance and autocorrelation for long on-policy simulation

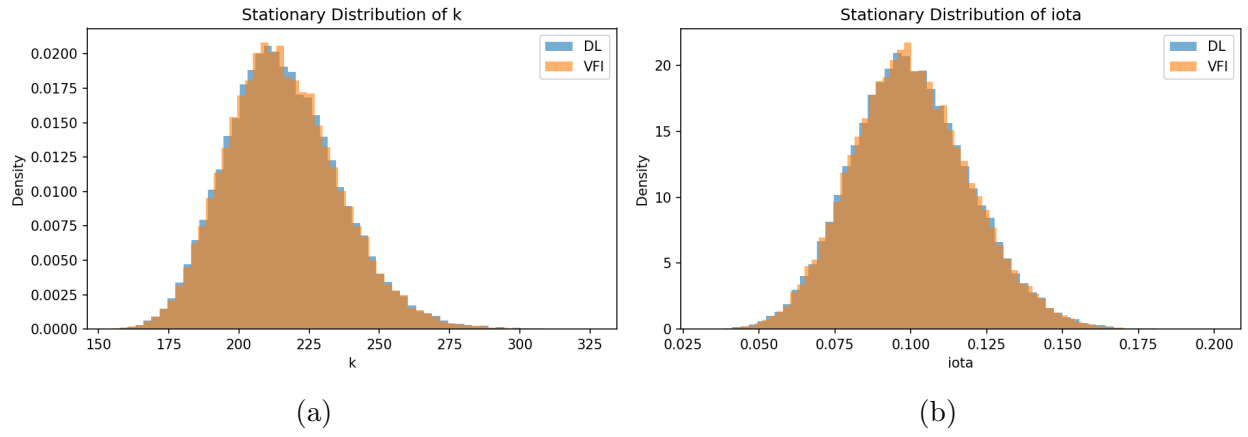


Figure 6: Stationary Distribution of k and ι

(DL–VFI). The DL and VFI regime maps are closely aligned in all three dimensions: regime means of ι , regime dispersion (std), and regime shares. The difference panels show only small residual deviations, indicating that the learned policy reproduces not only average investment behavior within each (k, z) regime but also its volatility and the frequency with which the economy visits each regime.(Fig.7).

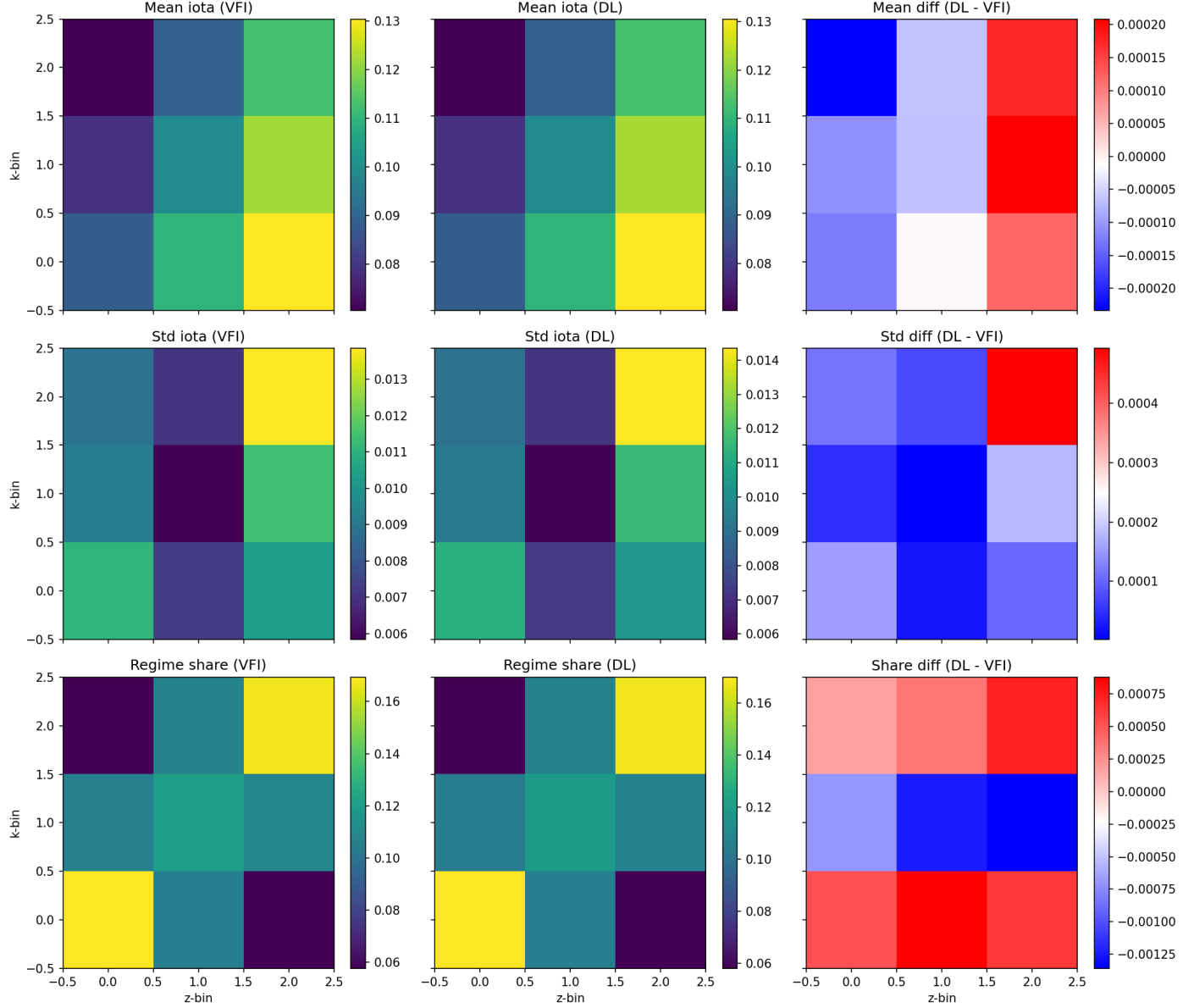


Figure 7: Regime behavior

3 Conclusion

In summary, this update has (i) clarified the theoretical foundations for using VFI/PFI as reliable low-dimensional benchmarks, via the contraction-mapping property of the discrete

Bellman operator under our modeling assumptions; (ii) explained why the Fischer–Burmeister (FB) construction in Maliar et al. (2021) cannot be mechanically transplanted into the limited-liability, endogenous-default environment of Section 3.6 in Strebulaev and Whited (2012); and (iii) provided a comprehensive numerical comparison between the DL and VFI solutions. The convergence diagnostics, policy and value surfaces, long-run ergodic moments, stationary distributions, and regime-by-regime statistics all indicate that the trained DL policy closely replicates the VFI benchmark, with residual discrepancies that are small and tightly centered. Taken together, these results support the conclusion that the deep-learning implementation delivers an alternative to the benchmark solution in the baseline setting and is therefore a credible tool to scale to higher-dimensional extensions where VFI becomes computationally infeasible.

Acknowledgement

I would like to thank the JP MLCOE team for providing insightful feedback and suggestions for codebase, further study and experiment.

References

- [1] Blackwell, D. (1965). Discounted dynamic programming. *Annals of Mathematical Statistics*, 36(1), 226–235.
- [2] Adda, J., Cooper, R. W. (2003). *Dynamic Economics: Quantitative Methods and Applications*. Cambridge, MA: The MIT Press.